# CSCI 232 Assignment 03 – C Pointers & Node Fundamentals

Points: 10

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## Instructions

1. **Fork** the repository on GitHub: `https://github.com/jpach-cs/232-ass03-template`
2. **Clone** your forked repository to your local machine.
3. Open `src/code.c` and `src/tests.c`.
4. Uncomment the author block at the top of `src/code.c` and fill in your name:

```
// char *AUTHOR_NAME       = "Your Name";
// char *AUTHOR_AUTHORSHIP = "I acknowledge that I have worked on this
// assignment independently...";
```

5. Work through each function in **two parts**:

### Part 1 — `src/code.c`

- Read the description and hint for each function.
- Implement the function body.
- Do **not** modify function signatures.

### Part 2 — `src/tests.c`

- Read the description for each test function.
- Implement the test body using `TEST_ASSERT_`.
- Use only **stack variables** — no `malloc`.
- Do **not** modify function names or signatures.

6. When all local tests pass — **commit** and **push** to your repository.
7. Go to the **Actions** tab on GitHub and verify the CI results.

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## Functions to implement — `src/code.c`

| Function | Description |
|---|---|
| `swap` | Swap values of two integers via pointers |
| `find_last_char` | Return pointer to last character before `\0` |
| `link_three` | Connect three nodes: `a -> b -> c -> NULL` |
| `remove_middle` | Reconnect `a -> c`, disconnect `b` |
| `remove_last` | Set `b->nextPtr = NULL` |

| Function | Description |
|---|---|
| `remove_first` | Set `a->nextPtr = NULL` |
| `swap_ptrs` | Swap two pointers using double pointers |
| `nullify` | Set pointer to NULL via double pointer |
| `assign_bytes` | Assign bytes 1..8 via cast pointer |
| `sum_chain` | Return sum of all values in chain |

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## Tests to implement — `src/tests.c`

| Test function | What to verify |
|---|---|
| `test_swap_basic` | Two different values are exchanged |
| `test_swap_equal` | Equal values remain unchanged |
| `test_find_last_basic` | Correct pointer to last character |
| `test_find_last_single` | Single character string |
| `test_find_last_empty` | Empty string returns NULL |
| `test_link_three_chain` | All three pointers correctly set |
| `test_remove_middle_reconnects` | `a->nextPtr == &c`, `b->nextPtr == NULL` |
| `test_remove_last_truncates` | `b->nextPtr == NULL` |
| `test_remove_first_isolates` | `a->nextPtr == NULL`, rest intact |
| `test_swap_ptrs_redirects` | Each pointer points to the other variable |
| `test_swap_ptrs_values_unchanged` | Original integer values not modified |
| `test_nullify_sets_null` | Pointer is NULL after call |
| `test_nullify_value_unchanged` | Original value not modified |
| `test_assign_bytes_first_and_last` | Byte 0 == 1, byte 7 == 8 |
| `test_assign_bytes_all` | Every byte i contains i+1 |
| `test_sum_chain_basic` | Correct sum for three nodes |
| `test_sum_chain_single` | Single node returns its value |
| `test_sum_chain_null` | NULL returns 0 |

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## Rules

- Work on **stack only** — no `malloc`, no `free`
- Do **not** modify function signatures in `code.c` or `tests.c`
- CI runs your `tests.c` together with instructor tests — both must pass

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## Grading

| Component | Points |
|---|---|
| All functions correctly implemented | 5 |
| All test functions correctly written | 5 |
| **Total** | **10** |

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## Author block — required

```
char *AUTHOR_NAME       = "Your Name";
char *AUTHOR_AUTHORSHIP = "..."
```